SOFTWARE PROJECT DOCUMENTATION

Requirements Engineering & Requirements Modeling

Project Name: Ease Travel's– Online Bus Ticket Booking System

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1. Software Requirements Specification (SRS)

1.1 Introduction

"Ease Travels" is a web-based online bus and 2D ticket booking system designed to make bus ticket booking easier, faster, and more organized. The system will provide a centralized platform where passengers can search for available buses according to their departure location, destination, and travel date.

Users will be able to view available buses, schedules, ticket prices, and available seats. They can select their preferred seat, enter passenger details, and complete the booking through the website. After successful booking, the system will generate an electronic ticket.

The system will also provide an administrative interface through which administrators can manage buses, routes, schedules, seats, users, and bookings.

1.2 Purpose

The main purpose of *Ease Travels* is to provide passengers with a convenient online platform for searching and booking bus tickets without having to visit bus terminals or contact bus AND 2D operators manually.

The system aims to:

- Make bus ticket booking easier and faster.
- Allow passengers to search for buses online.
- Provide information about bus schedules and ticket prices.

- Show available seats to passengers.
- Allow passengers to select their preferred seats.
- Allow users to enter and manage passenger details.
- Provide simulated online payment.
- Generate an electronic ticket after successful booking.
- Allow users to view their booking history.
- Allow users to cancel bookings.
- Help administrators manage buses, routes, schedules, seats, users, and bookings.

1.3 Scope

The scope of *Ease Travels* includes the development of a web-based platform for searching and booking bus tickets.

User-side features:

The system will allow passengers to:

- Register and log in.
- Search buses by departure location and destination.
- Select a travel date.
- View available buses.
- View bus schedules.
- View ticket prices.
- View available seats.
- Select a preferred seat.
- Enter passenger details.
- Book a bus ticket.
- Make a simulated online payment.
- Receive an electronic ticket.
- View current and previous bookings.
- Cancel a booking according to the system rules.

Administrator-side features:

- The administrator will be able to:
- Manage registered users.
- Add, update, and remove buses.
- Manage routes.
- Manage bus schedules.
- Manage seat availability.
- View and manage bookings.
- Update booking information.

- Maintain accurate bus and booking records.

Project Boundaries:

The initial version of Ease Travels will focus on online bus ticket searching and booking. The project will use *simulated online payment* rather than connecting to a real banking or payment gateway.

Figure 1 viewing buses at terminal



1.3 Survey:

Ease travel team conducted a survey to know the requirements of the bus companies and passenger through Qualitative Research .

The requirement gathering sub-team went to Bus and 2D Service Terminals to do inquiry of both the system and the problems related to manual booking system.

The main problem's faced by passengers were:

- The amount of distance they have to travel in order to book a ticket while their desired might be booked already or because of weather they might not be able to reach the terminal.
- They don't know the routes beforehand so they might face hardship when they reach their destination.
- Passengers weren't updated the time change so they missed their bus.

The main problems faced by the terminal operators:

- One terminal might already sell the same ticket again.so multiple people argue on a one seat.
- Record system being manual creates lots of issues since record might be damaged and they might even forget to note down a ticket then the passenger will have issues and complain.

Figure 2 survey with 2D and Bus Terminals



1.4 Stakeholders

Stakeholder 1: Passengers

Passengers are the main users of Ease Travels. They will use the website to search for buses, check schedules and prices, select seats, enter passenger information, and book tickets.

Stakeholder 2: Bus Operators

Bus operators will provide information about their buses, routes, schedules, seats, and ticket prices. They will benefit from organized booking records and better management of seat availability.

Stakeholder 3: Booking Staff

Booking staff may use the system to view and manage passenger bookings and assist with booking-related activities.

Stakeholder 4: System Administrator

The administrator will manage the overall system, including users, buses, routes, schedules, seats, and bookings.

Stakeholder 5: Development Team

The development team will design, develop, test, maintain, and deploy the Ease Travels system.

1.5 Functional Requirements

Functional requirements describe the functions that the Ease Travels system must provide.

FR-01: User Registration

The system shall allow new passengers to create an account by providing the required information.

FR-02: User Login

The system shall allow registered users to log in using their account credentials.

FR-03: Bus Search

The system shall allow users to search for buses by entering their departure location, destination, and travel date.

FR-04: View Available Buses

The system shall display available buses that match the user's search criteria.

FR-05: View Bus Schedule

The system shall allow users to view the departure time, arrival time, route, and other relevant schedule information.

FR-06: View Ticket Price

The system shall display the ticket price for each available bu

FR-07: View Seat Availability

The system shall display available and already-booked seats for a selected bus.

FR-08: Select Seat

The system shall allow users to select an available seat for their journey.

FR-09: Enter Passenger Details

The system shall allow users to enter required passenger information before confirming a booking.

FR-10: Book Bus Ticket

The system shall allow users to confirm and book a bus ticket after entering the required information.

FR-11: Simulated Online Payment

The system shall provide a simulated online payment process for completing the ticket booking.

FR-12: Generate E-Ticket

The system shall generate an electronic ticket after a successful booking.

FR-13: View Booking History

The system shall allow users to view their current and previous booking records.

FR-14: Cancel Booking
The system shall allow users to cancel a booking according to the defined cancellation rules.
FR-15: Manage Buses
The system shall allow administrators to add, update, and remove bus information.
FR-16: Manage Routes
The system shall allow administrators to add, update, and remove bus routes.
FR-17: Manage Schedules and Seats
The system shall allow administrators to manage bus schedules and seat availability.
FR-18: Manage Bookings
The system shall allow administrators to view and manage passenger bookings.

1.6 Non-Functional Requirements

Non-functional requirements describe how well the Ease Travels system should work.

NFR-01: Security

The system shall protect user account information and booking data from unauthorized access.
NFR-02: Performance

The system should provide bus search results within a reasonable amount of time.

NFR-03: Usability

The website shall have a simple and user-friendly interface so that passengers can easily search and book tickets.

NFR-04: Reliability

The system shall correctly store booking information and maintain accurate seat availability.

NFR-05: Availability

The website should be available to users whenever they need to search for or book bus tickets, subject to server and internet availability.

NFR-06: Data Integrity

The system shall maintain accurate and consistent information about users, buses, routes, schedules, seats, and bookings.

NFR-07: Maintainability

The system should be easy for developers and administrators to maintain and update. NFR-08: Compatibility

The website should work correctly on commonly used web browsers and different screen sizes.

NFR-09: Response Time

The system should respond to normal user actions, such as searching for buses or viewing bookings, without unnecessary delay.

NFR-10: Backup

Important user and booking information should be backed up to reduce the risk of data loss.

2. User Stories

US-01 – User Registration

As a passenger, I want to create an account so that I can use the booking services of Ease Travels.

US-02 – Search Bus

As a passenger, I want to search for buses by departure location, destination, and date so that I can find a suitable bus for my journey.

US-03 – View Bus Information

As a passenger, I want to view bus schedules, ticket prices, and available seats so that I can choose the bus that suits me.

US-04 – Select Seat

As a passenger, I want to select an available seat so that I can reserve my preferred seat.*

US-05 – Book Ticket

As a passenger, I want to book a bus ticket online so that I do not have to visit a bus terminal.

US-06 – Make Payment

As a passenger, I want to make a simulated online payment so that I can complete my ticket booking.

US-07 – Receive E-Ticket

As a passenger, I want to receive an electronic ticket after booking so that I have proof of my reservation.

US-08 – View Booking History

As a passenger, I want to view my booking history so that I can check my previous and upcoming journeys.

US-09 – Cancel Booking

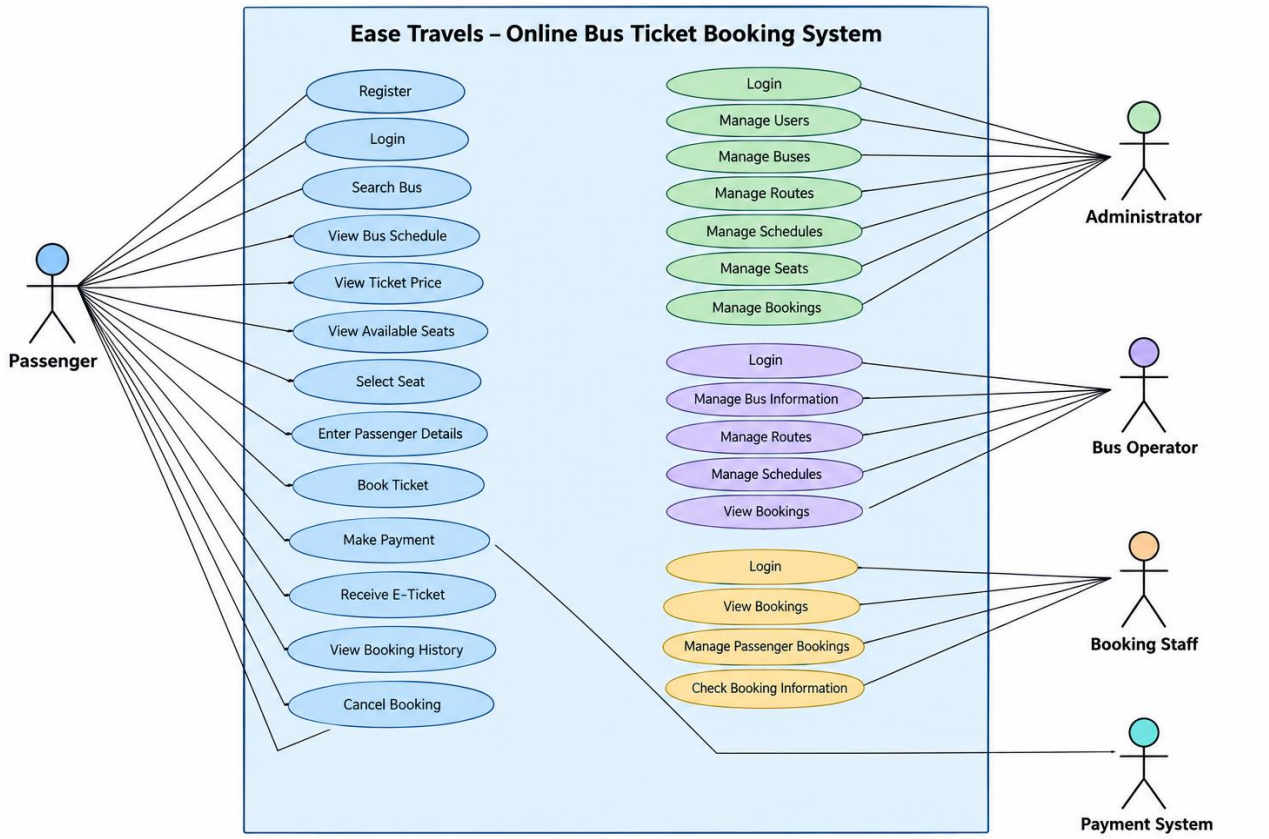
As a passenger, I want to cancel my booking so that I can cancel a journey when necessary.

US-10 – Manage System

As an administrator, I want to manage buses, routes, schedules, seats, users, and bookings so that the system information remains accurate and updated.

3. Use-Case Diagram

Actors: [List all actors, e.g., Student, Admin, Staff, Payment Service]



4. Acceptance Criteria

AC-01 – For User Story US-01: User Registration

Given: The passenger does not have an account.
When: The passenger enters valid registration information and submits the registration form.
Then: The system shall create the passenger's account and allow the passenger to log in.

AC-02 – For User Story US-02: Search Bus

Given: The passenger is on the bus search page.
When: The passenger enters the departure location, destination, and travel date.
Then: The system shall display available buses matching the entered search information.

AC-03 – For User Story US-03: View Bus Information

Given: Available buses are displayed.
When: The passenger selects a bus.

Then: The system shall display the bus schedule, ticket price, route, and available seat information.

AC-04 – For User Story US-04: Select Seat

Given: The passenger has selected an available bus.
When: The passenger selects an available seat.
Then: The system shall select the seat for the passenger and prevent already-booked seats from being selected.

AC-05 – For User Story US-05: Book Ticket

Given: The passenger has selected a bus and an available seat and entered valid passenger details.
When: The passenger confirms the booking.
Then: The system shall create the booking and display the booking information.

AC-06 – For User Story US-06: Make Payment

Given: The passenger has entered all required booking information.
When: The passenger completes the simulated payment process.
Then: The system shall display a successful payment status and continue the booking process.

AC-07 – For User Story US-07: Generate E-Ticket

Given: The passenger has successfully completed the booking and simulated payment.
When: The booking is confirmed.
Then: The system shall generate an electronic ticket containing relevant booking information.

AC-08 – For User Story US-08: View Booking History

Given: The passenger is logged into their account.
When: The passenger opens the booking history section.
Then: The system shall display the passenger's previous and upcoming bookings.

AC-09 – For User Story US-09: Cancel Booking

Given: The passenger has an existing eligible booking.
When: The passenger selects the cancel booking option.
Then: The system shall cancel the booking and update the booking status.

AC-10 – For User Story US-10: Manage System

Given: The administrator is logged into the administration panel.
When: The administrator adds, updates, or removes buses, routes, schedules, seats, users, or bookings.
Then: The system shall save the changes and display the updated information where applicable.

5. Quick Information Checklist

- Project name and basic project information
- Introduction
- Purpose
- Scope
- Stakeholders
- Functional requirements with IDs (FR-01, FR-02, ...)
- Non-functional requirements with IDs (NFR-01, NFR-02, ...)
- User stories
- Use-case actors and use cases
- Actual Use-Case Diagram
- Acceptance criteria for important user stories